# Kirlos Yousef

Senior Software Engineer | AI Application Engineering

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## SUMMARY

Senior software engineer with 5+ years shipping production software and leading engineering teams. Ships LLM features to production at Sellou: streaming chat, tool calling into native actions, and access-controlled search across iOS, web and AWS. Builds evaluated LLM systems in Python: schema-constrained extraction, hybrid retrieval with citations, and a bounded tool-calling agent exposed over MCP.

## TECHNICAL SKILLS

**LLM & agents:** OpenAI SDK, OpenRouter, Gemini, structured outputs, tool calling, LangGraph, MCP, human-in-the-loop approval, embeddings, reranking
**Evals & observability:** labelled eval sets, recall@k, MRR, prompt A/B tests, step tracing, token/cost tracking, PII redaction
**Backend & data:** Python, TypeScript, FastAPI, Pydantic, Hono, PostgreSQL, pgvector, DynamoDB, OpenSearch
**Cloud & delivery:** AWS Lambda, AppSync, CDK, Cloudflare Workers, Docker, GitHub Actions, pytest, Vitest, Ruff, Pyright
**Apps & clients:** Swift, SwiftUI, React, Next.js, GraphQL, SSE, OAuth

## PROFESSIONAL EXPERIENCE

**Sellou** — Utah, US (Remote) **02/2025 – Present**

**Lead Product Engineer – AI Applications** 08/2025 – Present

- Primary iOS and web engineer on Spookling, an AI workspace (~87% of iOS and ~76% of web commits)
- Streamed LLM responses over SSE and mapped model tool calls through typed boundaries to native EventKit calendar actions
- Shipped OpenSearch Serverless chat search with highlights and conversation-level access control
- Built workspace and DM chat APIs on AppSync, Lambda and DynamoDB with threads, mentions and file indexes
- Connected the assistant to Composio tools and Google/Buffer via OAuth; added assistant history and public booking APIs

**Senior iOS Engineer / iOS Tech Lead** 02/2025 – 08/2025

- Led the iOS team and mentored 2 engineers; introduced CI/CD that cut release cycle time by 50%
- Cut video playback startup 40% with AVKit/HLS and initial download size 20% with On-Demand Resources
- Built Plot My Calories end to end: meal photo to nutrition breakdown with OpenAI Vision and Swift Concurrency

**Iomob** — Barcelona, Spain (Remote) **02/2021 – 02/2025**

**iOS Team Lead** 02/2024 – 02/2025

- Led 5 iOS engineers delivering LNER and BrightBike at a crash-free rate above 99.9%
- Added Stripe tokenization to LNER, cutting checkout below 8 seconds; reduced trip-planning latency by 35%

**iOS Engineer** 02/2021 – 02/2024

- Built the mobility SDK behind Ford and Renfe integrations, cutting integration time from weeks to under 2 days
- Normalized data from 10+ providers; raised core SDK test coverage above 90%

## SELECTED AI PROJECTS

**AgentLookout** — AI coding-agent monitor (iOS/iPadOS + macOS) · App Store

- Built the iOS/iPadOS app, macOS connector and pairing relay that bring the open-source bot-crossing agent visualizer to phones, monitoring sessions across 7 tools incl. Claude Code, Codex and Cursor
- Shipped it solo to the App Store as AgentLookout: QR/code device pairing, live session status (running, waiting for input, completed) and a connector release pipeline

**Receipty** — Evaluated LLM extraction, RAG and agent (Python) · GitHub

- Built a FastAPI vision pipeline with strict JSON schemas, Pydantic validation and explicit review/failure outcomes; passed 60/60 labelled checks at p95 3.4 s and ~$0.0005 per receipt
- Built a labelled retrieval eval (69 queries; keyword, dense, hybrid and rerank strategies) behind answers that cite receipts or return not-found; recall@5 1.0 on the 64 answerable queries
- Built a LangGraph tool-calling agent with bounded steps and human approval for writes; exposed the same tools over an MCP server
- Added step tracing, token/cost logging and PII redaction; versioned prompts (v1 vs v2) are compared on the eval set, and CI enforces 178 tests and an 80% branch-coverage floor

**Revera AI** — AI voice recorder & transcriber (iOS/iPadOS) · App Store

- Built a Whisper/SFSpeech pipeline with offline queuing and interruption recovery; cut upload data ~82% via 16 kHz mono conversion
- Put OpenAI behind a Cloudflare Worker with allowlisted routes and server-side usage limits, so no API key ships in the app

## EDUCATION

B.S. Computer Science, University of Debrecen, Hungary · 2017 – 2020 · Graduated first in cohort, fully funded scholarship